

EXERCISE 1

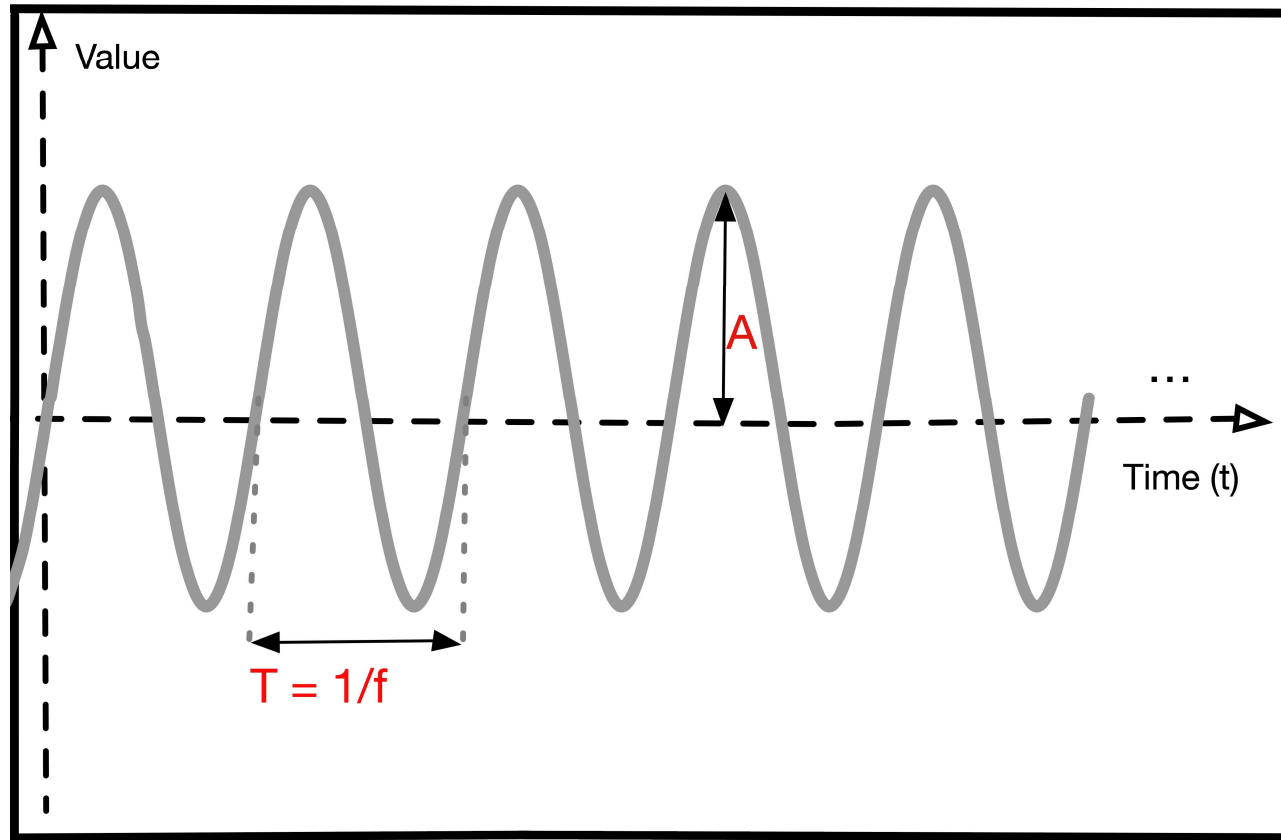
Multimedia Databases (Exercises)

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1: Representing an Analog Signal



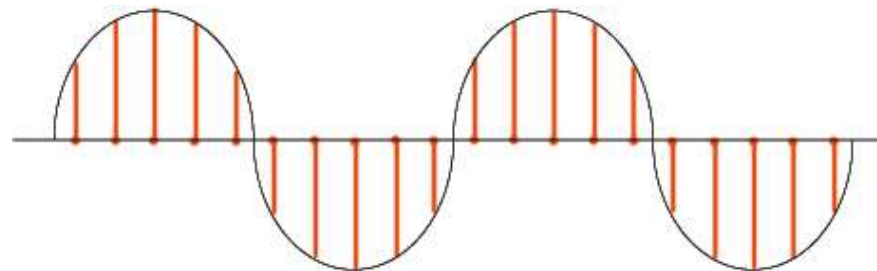
A is the amplitude of the sine curve (the peak deviation of the function from zero.)

T is the period of the sine curve (length of one cycle of the curve).

f is the frequency (number of cycles per second)

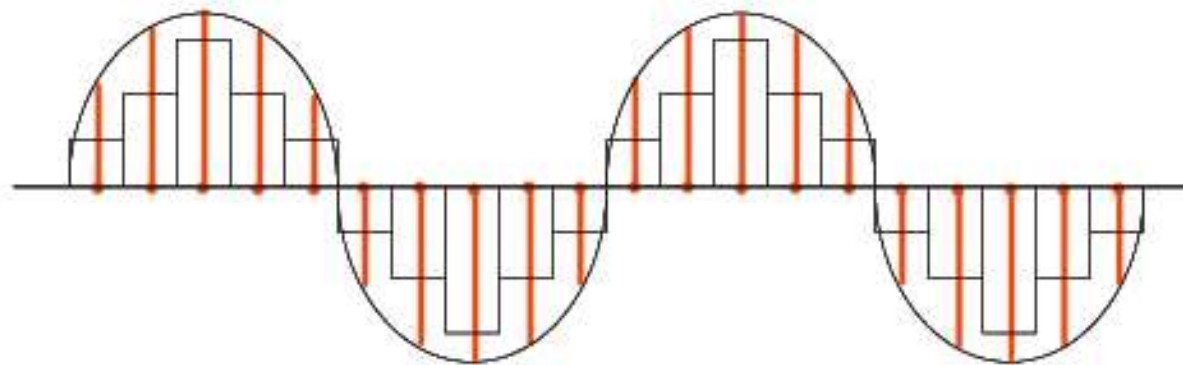
1: PCM Sampling

- With sampling, a fixed grid of measuring points of equal distance Δt is defined on the axis over which the analog signal changes. The current value(s) of the signal at the measuring points of this grid is called a sample.
 - The density of the measured values is called sampling rate.
 - The fixed grid can be points in time (example audio signal) or spatial dimensions (example pixels).
 - The sampling rate specifies how accurately the original signal can be reconstructed.



1: PCM Quantization

- The quantization corresponds to the conversion of the measured values obtained during discretization to a discrete, countable value range (usually in the binary system).
 - Resolution: bits per sample (bit resolution)
 - The accuracy depends on the number of bits per measured value



Quantization Error

- Maximum Quantization Error $Q = \frac{\Delta x}{2^N \cdot 2}$

Δx is the range of the signal, N is the number of bits

- The range of our signal $\Delta x = 10 - (-10) \text{ V} = 20 \text{ V}$

Therefore,

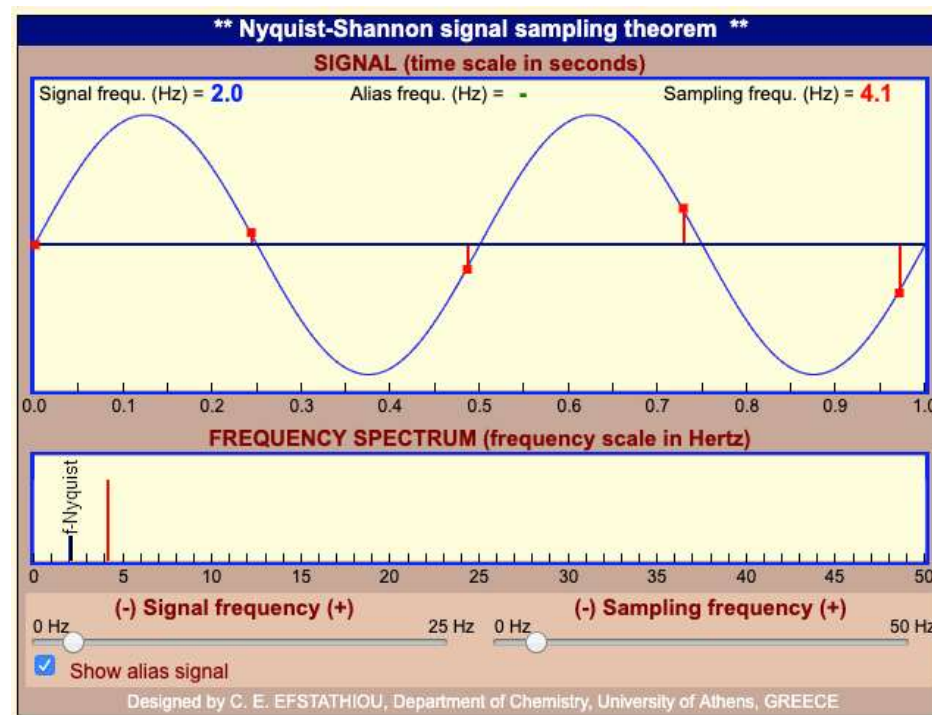
$$Q = 20 / (2^N \cdot 2) = 20 / 64 = 0.3125 \text{ V}$$



1: Nyquist-Shannon Sampling Theorem.

If a function (signal) with the highest occurring frequency f_g is sampled at a sampling rate f_s , so that $f_s > 2 * f_g$, then this function can be reconstructed from the sampled values without losing the underlying information.

Developed by Harry Nyquist (1928) and proved by Claude Shannon.



Applet: http://195.134.76.37/applets/AppletNyquist/Applet_Nyquist2.html



1: Nyquist-Shannon Sampling Theorem.

- Given,

$$f(x) = \sin(0.8\pi x) + \sin(1.5\pi x) + \sin(3\pi x)$$

- We must first determine the individual frequency components

$$f(x) = \sin(2\pi f_1 x) + \sin(2\pi f_2 x) + \sin(2\pi f_3 x)$$

- Therefore,

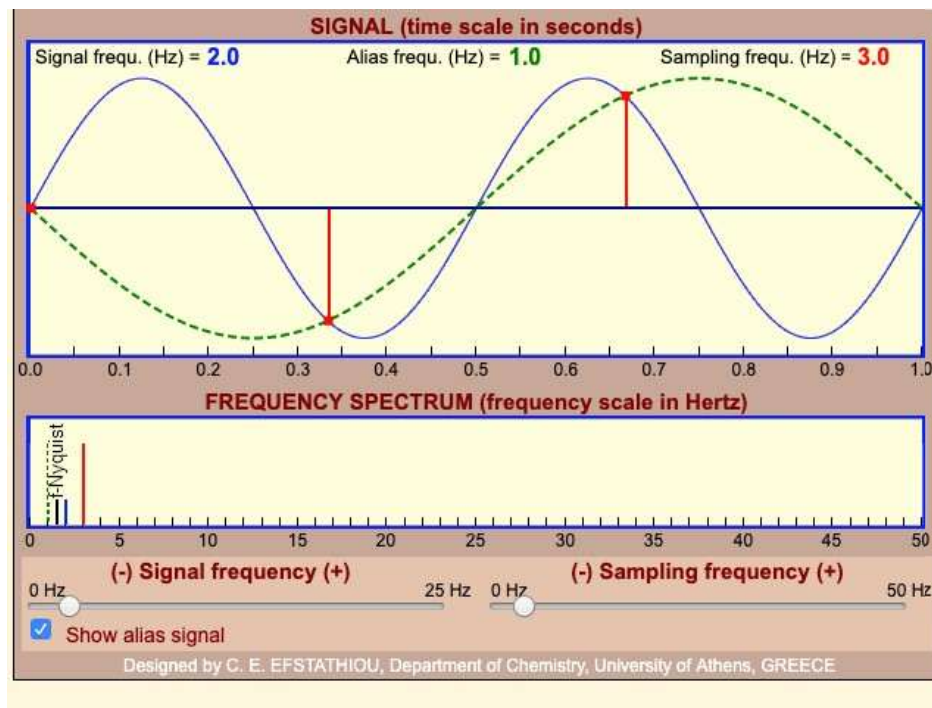
$$f_1 = 0.4 \text{ Hz}, f_2 = 0.75 \text{ Hz}, f_3 = 1.5 \text{ Hz} \text{ (} f_3 \text{ is the highest frequency component)}$$

- Thus, the minimum sample rate $f_s > 2 \cdot f_3 \Rightarrow f_s > 3 \text{ Hz}$



1: Aliasing

Aliasing arises when a signal is discretely sampled at a rate that is insufficient to capture the changes in the signal.



No Anti-Aliasing

Anti-Aliasing

[Source: greenprinter.com](http://greenprinter.com) What is anti-aliasing?

Aliasing due to a low sampling rate (less than twice the highest frequency component)

2: Structured and Unstructured Data

Structured	Unstructured	Semi-Structured
• Predefined schema • Referred to as quantitative data.	• It has no pre-defined model. • Collecting, processing, and analyzing unstructured data represents a significant challenge.	• Does not obey the formal structure of data models associated with relational databases or other forms of data tables. • Contain tags or other markers to separate semantic elements and enforce hierarchies of records and fields within the data.
Data that fits within fixed fields and columns in relational databases. Names, addresses, location etc.	Text, video, audio	XML, JSON

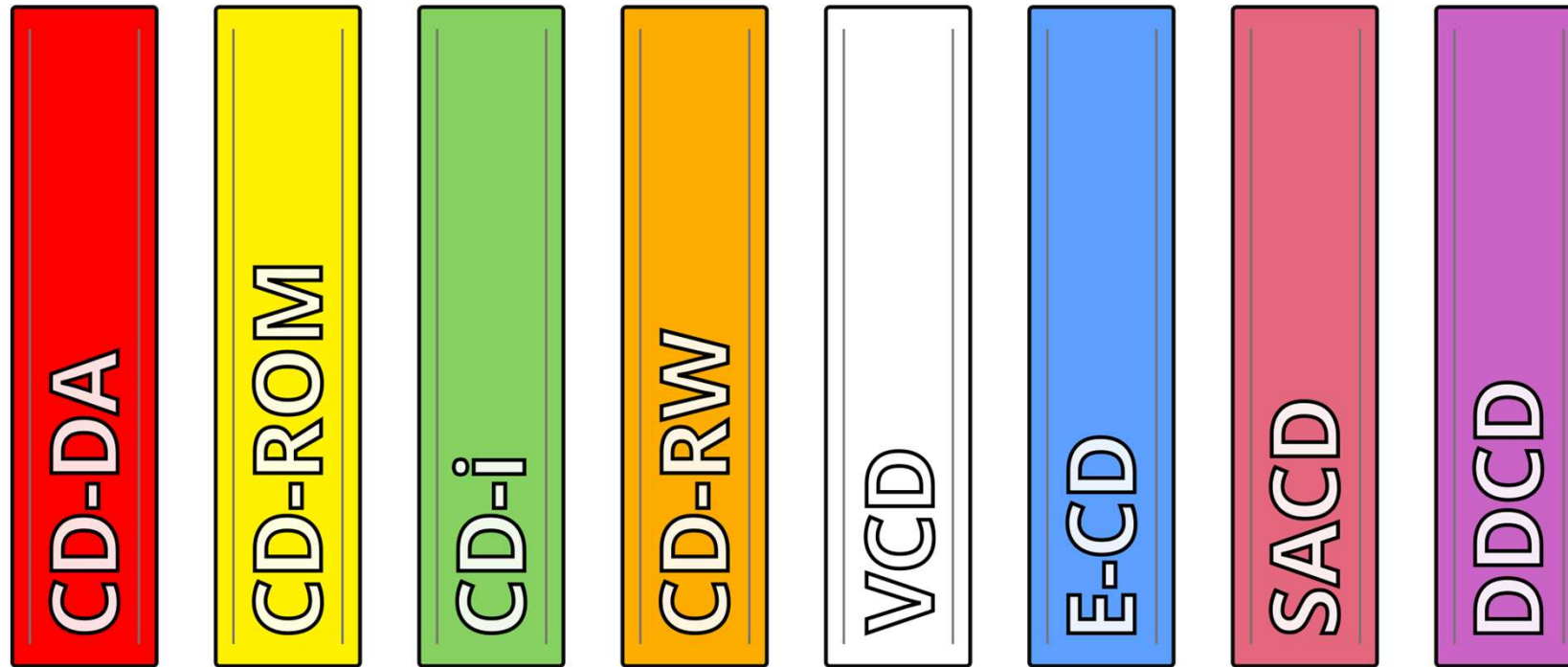


2: Structured and Unstructured Data

Structured	Unstructured	Semi-Structured
Can be implemented/stored in a relational DB	• Can't reside in a traditional row-column DB	• Can't reside in a traditional row-column DB
• Standard SQL querying • Hits exact matching	Metadata is needed. Structured? • $Y \Rightarrow$ exact matching (e.g. videos by an author) • $N \Rightarrow$ Information Retrieval methods • Fuzzy matching e.g. Similarity-based comparison	Typically requires either a structured query such as XPath, or a keyword query that does not take structure into account.



3: Rainbow Books



By Fabien Sanglard - https://fabiensanglard.net/doom_psx/rainbow_books.svg, Public Domain

3: Rainbow Books

Book	Media	Additions
CD-DA	Continuous Audio	24 bit 79mins Audio
CD-ROM	File Storage and Error S.	Non Continuous Files
CD-i	Interactive Elements	Embedding of dif. Types
CD-M/R/RW	Dynamic Filestorage	Multi session
Video-CD	Mix of cont. and interactive	High fidelity video formats
E-CD	Cont. Audio w. File storage	Compatibility w. Multi S.
SuperAudioCD	Cont. Audio w. copy prot.	Policy Enforcements
DDCD	File Storage ...	new Storage Structure

... and many more additional details, refer to [Compact Disc Rainbow Books | Museum of Obsolete Media](https://obsoletedia.org/audio/compact-disc-rainbow-books/)
<https://obsoletedia.org/audio/compact-disc-rainbow-books/>

